

Beginner Installation Guide for bib_maker

This guide explains exactly how to install and use the **bib_maker** repository on Windows, Linux, and macOS, even if you have little or no programming experience.

What is bib_maker?

bib_maker is a simple Python script that creates a .bib bibliography file from a list of DOIs or from an existing .bib file.

The repository contains:

- bib_maker.py → the main script
- journal_abbreviations.csv → abbreviation database
- examples/ → example input files

The script depends on a few Python libraries:

- pandas
- pybtex
- requests

Recommended Installation Method

The easiest and safest method is to install Python using Miniconda. This avoids many common beginner problems.

You can also use normal Python + pip if you prefer.

WINDOWS INSTALLATION (Recommended Method)

STEP 1 – Install Miniconda

1. Open your browser.
2. Go to:
<https://www.anaconda.com/download/success>
3. Download Miniconda for Windows.
4. Run the installer.
5. During installation:
 - Choose “Just Me”
 - Keep default settings
 - Allow Miniconda to add shortcuts

STEP 2 – Open the Miniconda Prompt

1. Open the Windows Start Menu.
2. Search for:
Miniconda Prompt
3. Open it.

STEP 3 – Create a clean Python environment

Copy and paste this command:

```
conda create -n bibmaker python=3.11 -y
```

Press ENTER.

STEP 4 – Activate the environment

Copy and paste:

```
conda activate bibmaker
```

You should now see:
(bibmaker)

at the beginning of the terminal line.

STEP 5 – Install required libraries

Copy and paste:

```
pip install pandas pybtex==0.26.0 requests
```

STEP 6 – Extract the repository ZIP

1. Right-click the downloaded ZIP file.
2. Choose "Extract All".
3. Put it somewhere easy to find, for example:

```
C:\Users\YOUR_NAME\Documents\bib_maker
```

STEP 7 – Move into the repository folder

Example:

```
cd C:\Users\YOUR_NAME\Documents\bib_maker
```

STEP 8 – Test that it works

Run:

```
python bib_maker.py --help
```

If everything is working, you should see the usage instructions.

LINUX INSTALLATION

STEP 1 – Install Miniconda

Open a terminal and download Miniconda:

```
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
```

Run the installer:

```
bash Miniconda3-latest-Linux-x86_64.sh
```

Restart the terminal after installation.

STEP 2 – Create an environment

```
conda create -n bibmaker python=3.11 -y
```

STEP 3 – Activate it

```
conda activate bibmaker
```

STEP 4 – Install dependencies

```
pip install pandas pybtex==0.26.0 requests
```

STEP 5 – Extract the repository ZIP

```
unzip bib_maker-main.zip
```

STEP 6 – Move into the folder

```
cd bib_maker-main
```

STEP 7 – Test the installation

```
python3 bib_maker.py --help
```

macOS INSTALLATION

STEP 1 – Install Miniconda

Download Miniconda for macOS from:

<https://www.anaconda.com/download/success>

Run the installer.

STEP 2 – Open Terminal

Press CMD + SPACE and search for:
Terminal

STEP 3 – Create an environment

```
conda create -n bibmaker python=3.11 -y
```

STEP 4 – Activate the environment

```
conda activate bibmaker
```

STEP 5 – Install dependencies

```
pip install pandas pybtex==0.26.0 requests
```

STEP 6 – Extract the ZIP and enter the folder

```
cd path/to/bib_maker-main
```

STEP 7 – Test the installation

```
python3 bib_maker.py --help
```

How to Use bib_maker

The basic command is:

```
python bib_maker.py input_file output_file.bib
```

Example:

```
python bib_maker.py my_dois.txt references.bib
```

The input file can contain:

```
10.1103/PhysRevLett.116.061102
```

or:

```
einstein1905 10.1002/andp.19053221004
```

You can also give it a .bbl file directly.

Useful Options

-o
Overwrite existing output files.

-v
Verbose mode. Shows progress information.

-f
Force processing even if some DOIs are missing.

Example:

```
python bib_maker.py -o -v input.txt output.bib
```

Common Beginner Problems

PROBLEM:
'python' is not recognized

SOLUTION:
Python or Conda is not installed correctly.
Restart the terminal after installing Miniconda.

PROBLEM:

ModuleNotFoundError

SOLUTION:

You forgot to install dependencies.

Run:

```
pip install pandas pybtex==0.26.0 requests
```

PROBLEM:

UnicodeDecodeError on Windows

SOLUTION:

Open files using UTF-8 encoding or run the terminal in UTF-8 mode.

You can also try:

```
set PYTHONUTF8=1
```

before running the script.

PROBLEM:

conda command not found

SOLUTION:

Restart the terminal or reinstall Miniconda.

Recommended Folder Layout

A clean setup might look like this:

```
Documents/  
■■■ bib_maker/  
    ■■■ bib_maker.py  
    ■■■ journal_abbreviations.csv  
    ■■■ examples/  
    ■■■ input.txt  
    ■■■ output.bib
```

Final Notes

This repository is currently a single standalone Python script rather than a packaged Python application.

That means:

- there is no installer
- there is no setup.py
- there is no pip package

You simply:

1. install Python
2. install dependencies
3. run the script manually

The repository was written using pybtex version 0.26.0, so this guide installs that exact version for maximum compatibility.